

Mathematics-I
GTU#(3110014)



Dedicated Faculty, Committed Education
Darshan
Institute of Engineering & Technology

Unit:1.1

Indeterminate Forms



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OUTLINE

What is Indeterminate Forms?

L'Hospital's Rule

Method-1 $\rightarrow \left(\frac{0}{0} \text{ Form} \right)$

Method-2 $\rightarrow \left(\frac{\infty}{\infty} \text{ Form} \right)$

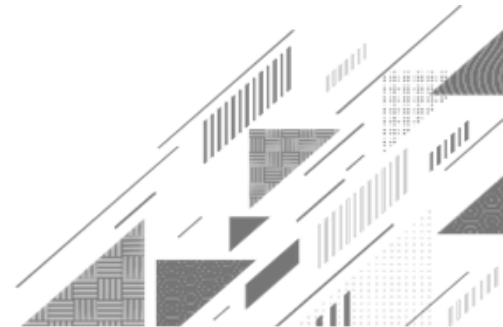
OUTLINE

Method-3 ➡ **($0 \times \infty$ Form)**

Method-4 ➡ **($\infty - \infty$ Form)**

Method-5 ➡ **($0^0, \infty^0$ & 1^∞ Form)**

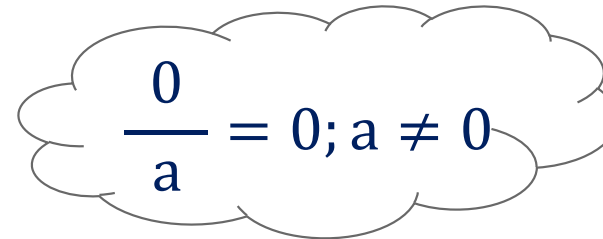
What is Indeterminate Form?



What is Indeterminate Forms?

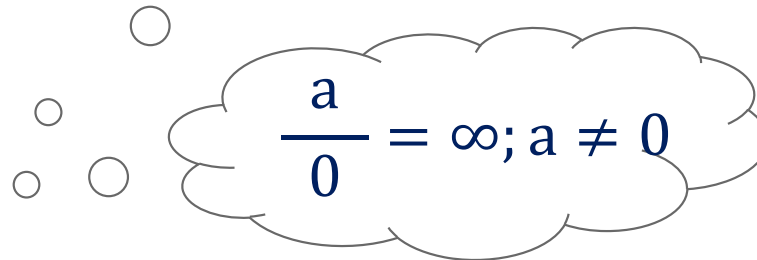
- Suppose we want to find $\lim_{x \rightarrow 0} \frac{x^2}{5}$

$$\text{Then, } \lim_{x \rightarrow 0} \frac{x^2}{5} = \frac{0}{5} = 0$$


$$\frac{0}{a} = 0; a \neq 0$$

- Now, if we want to find $\lim_{x \rightarrow 0} \frac{\sin x}{x}$

$$\text{Then, } \lim_{x \rightarrow 0} \frac{\sin x}{x} = \frac{\sin 0}{0} = \frac{0}{0}$$


$$\frac{a}{0} = \infty; a \neq 0$$



The form $\frac{0}{0}$ is called Indeterminate Form

What is Indeterminate Forms?

► Suppose we want to find $\lim_{x \rightarrow \infty} \frac{e^x}{x^2}$

Then,

$$\lim_{x \rightarrow \infty} \frac{e^x}{x^2} = \frac{e^\infty}{\infty}$$
$$= \frac{\infty}{\infty}$$



The form $\frac{\infty}{\infty}$ is called Indeterminate Form

Types of Indeterminate Forms

$$\infty - \infty$$

$$0 \times \infty$$

$$0^0$$

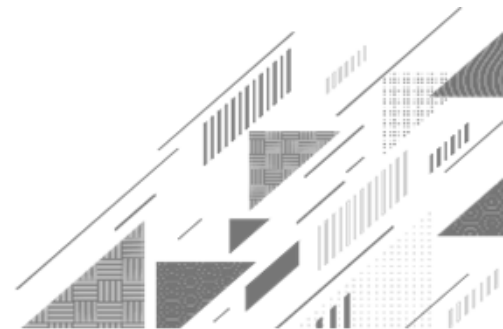
$$\frac{\infty}{\infty}$$

$$1^\infty$$

$$\frac{0}{0}$$

$$\infty^0$$

L'Hospital's Rule



L'Hospital's Rule

- ▶ L'Hospital's rule provides a technique to evaluate limits of indeterminate forms.
- ▶ This rule converts an indeterminate forms to an expression which helps to evaluate limit easily.
- ▶ If

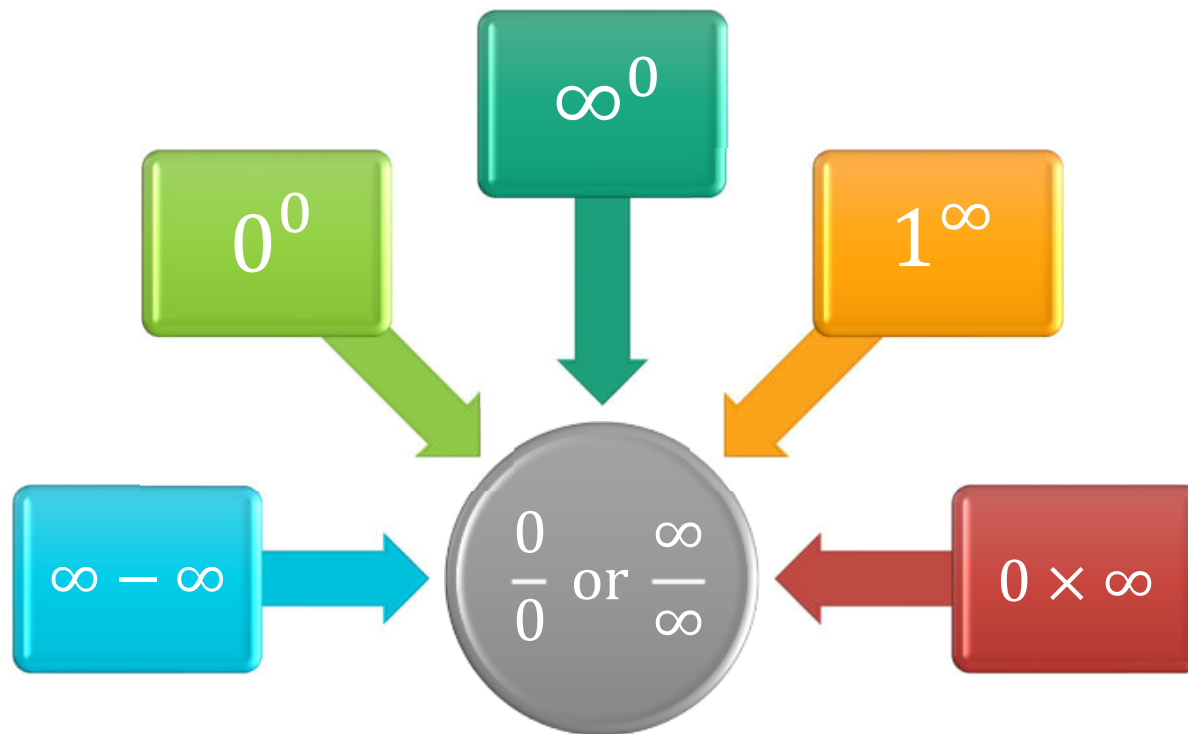
$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{f(a)}{g(a)} = \frac{0}{0} \text{ or } \frac{\infty}{\infty}$$

then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} \text{ provided the later limit exists.}$$

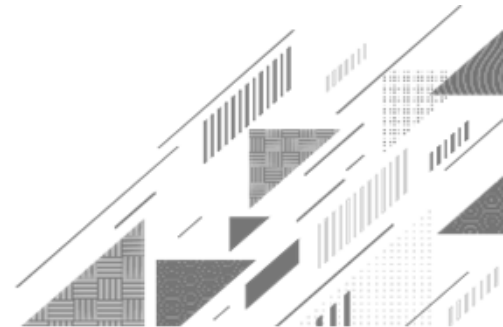
L'Hospital's Rule

- ▶ L'Hospital's Rule is applicable for $\frac{0}{0}$ or $\frac{\infty}{\infty}$ type indeterminate forms only.
- ▶ To find the limits of other indeterminate forms, we will convert them into $\frac{0}{0}$ or $\frac{\infty}{\infty}$.





Method:1 $\rightarrow \frac{0}{0}$ Form



Method:1 $\rightarrow$ $\left(\frac{0}{0} \text{ Form}\right)$

Ide Apply L'Hospital's Rule. rm.

Example:1 Evaluate $\lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2}$.

Solution:

$$\lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2} \rightarrow \frac{e^0 - 1 - 0}{0} = \frac{1 - 1}{0} \left(\frac{0}{0}\right)$$

$$= \lim_{x \rightarrow 0} \frac{e^x - 1}{2x}$$

$$\frac{d}{dx} (e^x - 1 - x) = e^x - 0 - 1$$

$$\frac{d}{dx} (x^2) = 2x$$

Method:1 $\Rightarrow$ Example:1(Continue) Apply the limit.

$$= \lim_{x \rightarrow 0} \frac{e^x - 0 - 1}{2x} = \lim_{x \rightarrow 0} \frac{e^x - 1}{2x} \longrightarrow \frac{e^0 - 1}{2 \cdot 0} = \frac{1 - 1}{0} \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 0} \frac{e^x - 0}{2}$$

$$= \frac{e^0}{2}$$

$$= \frac{1}{2}$$

By L'Hospital's Rule

If it again reduces to the form $\frac{0}{0}$ or $\frac{\infty}{\infty}$, apply L'Hospital's Rule.

Method:1 $\rightarrow$ $\left(\frac{0}{0} \text{ Form}\right)$

Example:2 Find $\lim_{x \rightarrow 0} \frac{\sin x}{x}$

$\tan x$

Solution:

$$\text{Here, } \lim_{x \rightarrow 0} \frac{\sin x}{x} \rightarrow \frac{\sin 0}{0} \left(\frac{0}{0}\right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{\cos x}{1}\right) \quad \therefore \text{L'Hospital's Rule}$$

$$= \frac{\cos 0}{1} = 1$$

$$\text{Here, } \lim_{x \rightarrow 0} \frac{\tan x}{x} \rightarrow \frac{\tan 0}{0} \left(\frac{0}{0}\right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{\sec^2 x}{1}\right) \quad \therefore \text{L'Hospital's Rule}$$

$$= \frac{\sec^2 0}{1}$$

$$= 1$$

Method:1 $\rightarrow$ $\left(\frac{0}{0} \text{ Form}\right)$

Apply the limit.

Example:3

Evaluate $\lim_{x \rightarrow \frac{\pi}{2}} \left(\frac{2x - \pi}{\cos x} \right) \left(\frac{0}{0} \right)$

Solution:

$$\lim_{x \rightarrow \frac{\pi}{2}} \left(\frac{2x - \pi}{\cos x} \right)$$

$$= \lim_{x \rightarrow \frac{\pi}{2}} \left(\frac{2 - 0}{-\sin x} \right)$$

$$= \left(-\frac{2}{\sin \frac{\pi}{2}} \right) = -2$$

By L'Hospital's Rule

$$\sin \frac{\pi}{2} = 1$$

Remember

$$\rightarrow \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)$$

$= 1$

$$\rightarrow \lim_{x \rightarrow 0} \left(\frac{\sin^2 x}{x^2} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^2 = \left(\lim_{x \rightarrow 0} \frac{\sin x}{x} \right)^2 = 1^2 = 1$$

$$\rightarrow \lim_{x \rightarrow 0} \left\{ \frac{\sin^8 3x}{(3x)^8} \right\} = \left(\lim_{3x \rightarrow 0} \frac{\sin 3x}{3x} \right)^8 = 1^8 = 1$$

Note that, $(\sin x)^2 = \sin^2 x$

but, $(\sin x)^2 \neq \sin x^2$

Method:1 $\rightarrow$ $\left(\frac{0}{0} \text{ Form}\right)$

Example:4 Evaluate $\lim_{x \rightarrow 0} \left\{ \frac{(\sin x)^2 - x^2}{x^2 (\sin x)^2} \right\}$

Solution:

$$\text{Here, } \lim_{x \rightarrow 0} \left\{ \frac{(\sin x)^2 - x^2}{x^2 (\sin x)^2} \right\}$$

$$= \lim_{x \rightarrow 0} \left\{ \frac{(\sin x)^2 - x^2}{x^2 \cdot \frac{x^2 (\sin x)^2}{x^2}} \right\}$$

$$= \lim_{x \rightarrow 0} \left\{ \frac{(\sin x)^2 - x^2}{x^4 \cdot \left(\frac{\sin^2 x}{x^2} \right)} \right\}$$

$$= \lim_{x \rightarrow 0} \left\{ \frac{(\sin x)^2 - x^2}{x^4 \cdot \left(\frac{\sin x}{x} \right)^2} \right\}$$

$$= \lim_{x \rightarrow 0} \left(\frac{\sin^2 x - x^2}{x^4} \right) \left(\frac{0}{0} \right)$$

$$\lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right) = 1$$

Method:1 $\Rightarrow$ Example:4(Continue)

Apply L'Hospital's Rule

$$= \lim_{x \rightarrow 0} \left(\frac{\sin^2 x - x^2}{x^4} \right) \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{2 \sin x \cos x - 2x}{4x^3} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{\sin 2x - 2x}{4x^3} \right) \longrightarrow \left(\frac{\sin 0 - 2(0)}{4 \cdot 0} \right) \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{2(\cos 2x) - 2}{12x^2} \right)$$

$$\frac{d}{dx} (\sin 2x) = 2 \cdot (\cos 2x)$$

$$\frac{d}{dx} (\sin^2 x) = 2 \sin x \cos x$$

$$2 \sin x \cos x = \sin 2x$$

By L'Hospital's Rule